

Quantum Mechanics Essentials

A compact reference for the introductory concepts used in the course

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Big picture. Quantum mechanics describes a state with a wavefunction (or state vector), represents measurable quantities with operators, and obtains special states by solving eigenvalue equations. The notation below is the language we use throughout DFT and ab-initio molecular dynamics.

1. Quantum states, operators, and eigenvalue equations

A quantum state is represented abstractly by a ket

$$|\Psi\rangle.$$

In coordinate representation, the same state is described by a wavefunction

$$\Psi(x) = \langle x | \Psi \rangle.$$

An operator acts on a state. For example, the Hamiltonian $\hat{H}$ is the energy operator. The time-independent Schrödinger equation is

$$\hat{H}|\Psi_n\rangle = E_n|\Psi_n\rangle.$$

If applying $\hat{H}$ to $|\Psi_n\rangle$ returns the same state multiplied by a number, then:

- $|\Psi_n\rangle$ is an eigenstate (or eigenfunction),
- E_n is the corresponding eigenvalue.

Interpretation: an eigenstate is a state with a definite value of the observable represented by the operator. For the Hamiltonian, the eigenvalue is an allowed energy.

2. Bra-ket notation and complex conjugation

The ket can be visualized as a column vector,

$$|\Psi\rangle = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}.$$

The corresponding bra is its Hermitian conjugate:

$$\langle\Psi| = |\Psi\rangle^\dagger,$$

where $\dagger$ means

transpose + complex conjugate.

Therefore,

$$\langle\Psi| = (c_1^* \quad c_2^* \quad c_3^*).$$

What is a complex conjugate? If

$$z = a + ib, \quad \text{then} \quad z^* = a - ib.$$

We simply reverse the sign of the imaginary part. For a wavefunction, Ψ^* means taking the complex conjugate at every point. A useful identity is

$$z^* z = |z|^2 = a^2 + b^2 \geq 0, \quad \text{and analogously} \quad \boxed{\Psi^* \Psi = |\Psi|^2}.$$

This is real and non-negative, which is why $|\Psi|^2$ can be interpreted probabilistically.

3. Inner products, normalization, and orthonormality

Combining a bra and a ket gives an **inner product**:

$$\langle \Phi | \Psi \rangle.$$

It is a number that measures the overlap between two states. In coordinate representation,

$$\langle \Phi | \Psi \rangle = \int \Phi^*(x) \Psi(x) dx.$$

For a many-electron wavefunction the integration extends over all electronic coordinates. A normalized state satisfies

$$\langle \Psi | \Psi \rangle = \int |\Psi(x)|^2 dx = 1.$$

Two states are orthogonal when

$$\langle \Psi_m | \Psi_n \rangle = 0 \quad (m \neq n).$$

A set of normalized, mutually orthogonal states is **orthonormal**:

$$\langle \Psi_m | \Psi_n \rangle = \delta_{mn},$$

with $\delta_{mn} = 1$ for $m = n$ and 0 otherwise.

4. Hermitian operators

An operator is Hermitian if it is equal to its own Hermitian conjugate:

$$\hat{A}^\dagger = \hat{A}.$$

In quantum mechanics, observables are represented by Hermitian (more precisely, self-adjoint) operators. Two consequences are especially important:

1. their eigenvalues are real;
2. eigenstates belonging to different eigenvalues are orthogonal.

For matrix elements, Hermiticity means

$$\langle \Phi | \hat{A} | \Psi \rangle = \langle \hat{A} \Phi | \Psi \rangle.$$

For the Hamiltonian, this is what allows us to **move $\hat{H}$ from the ket side to the bra side** in many derivations.

Why are the eigenvalues real?

Start from the eigenvalue equation:

$$\hat{H} | \Psi_n \rangle = E_n | \Psi_n \rangle.$$

Take the Hermitian conjugate of the **whole equation**:

$$(\hat{H} | \Psi_n \rangle)^\dagger = (E_n | \Psi_n \rangle)^\dagger.$$

Using $(AB)^\dagger = B^\dagger A^\dagger$ and the fact that a scalar becomes its complex conjugate,

$$\langle \Psi_n | \hat{H}^\dagger = E_n^* \langle \Psi_n |.$$

Because $\hat{H}$ is Hermitian, $\hat{H}^\dagger = \hat{H}$:

$$\langle \Psi_n | \hat{H} = E_n^* \langle \Psi_n |.$$

Multiply this equation on the right by $| \Psi_n \rangle$:

$$\langle \Psi_n | \hat{H} | \Psi_n \rangle = E_n^* \langle \Psi_n | \Psi_n \rangle.$$

From the original equation, multiply on the left by $\langle \Psi_n |$:

$$\langle \Psi_n | \hat{H} | \Psi_n \rangle = E_n \langle \Psi_n | \Psi_n \rangle.$$

The left-hand side is the same in both expressions, therefore

$$E_n \langle \Psi_n | \Psi_n \rangle = E_n^* \langle \Psi_n | \Psi_n \rangle.$$

For a normalized state, $\langle \Psi_n | \Psi_n \rangle = 1$, so

$$E_n = E_n^* \Rightarrow E_n \in \mathbb{R}.$$

Take-home message: Hermiticity guarantees that measured values such as energies are real numbers.

5. Why are eigenstates orthogonal?

Take two eigenstates of the same Hermitian Hamiltonian:

$$\hat{H}|\Psi_n\rangle = E_n|\Psi_n\rangle, \quad \hat{H}|\Psi_m\rangle = E_m|\Psi_m\rangle.$$

Step 1 – let $\hat{H}$ act on $|\Psi_n\rangle$.

Multiply the first eigenvalue equation on the left by $\langle\Psi_m|$:

$$\langle\Psi_m|\hat{H}|\Psi_n\rangle = E_n\langle\Psi_m|\Psi_n\rangle.$$

Step 2 – use Hermiticity to move $\hat{H}$ from the ket side to the bra side.

For a Hermitian Hamiltonian, $\hat{H}^\dagger = \hat{H}$, so

$$\langle\Psi_m|\hat{H}|\Psi_n\rangle = \langle\hat{H}\Psi_m|\Psi_n\rangle.$$

Now use the second eigenvalue equation, $\hat{H}|\Psi_m\rangle = E_m|\Psi_m\rangle$. Since E_m is real for a Hermitian operator,

$$\langle\hat{H}\Psi_m|\Psi_n\rangle = E_m\langle\Psi_m|\Psi_n\rangle.$$

Therefore the same matrix element has two expressions:

$$E_n\langle\Psi_m|\Psi_n\rangle = E_m\langle\Psi_m|\Psi_n\rangle,$$

which gives

$$(E_n - E_m)\langle\Psi_m|\Psi_n\rangle = 0.$$

Case 1: $m \neq n$ and $E_m \neq E_n$.

Then $E_n - E_m \neq 0$, so necessarily

$$\langle\Psi_m|\Psi_n\rangle = 0.$$

The two eigenstates are orthogonal.

Case 2: $m = n$.

The equation above becomes $0 = 0$ and tells us nothing about the length of the state. We therefore normalize each eigenstate by choosing its overall scale such that

$$\langle\Psi_n|\Psi_n\rangle = 1.$$

Combining orthogonality for different states with normalization for each state gives

$$\langle\Psi_m|\Psi_n\rangle = \delta_{mn},$$

which is orthonormality.

Degeneracy: if $m \neq n$ but $E_m = E_n$, the argument above does not force the states to be orthogonal. Within a degenerate subspace, we can choose orthogonal linear combinations and then normalize them.

6. Expectation values

For a normalized state $|\Psi\rangle$, the expectation value of an observable $\hat{A}$ is

$$\langle A \rangle = \langle\Psi|\hat{A}|\Psi\rangle.$$

In coordinate representation,

$$\langle A \rangle = \int \Psi^*(x) \hat{A} \Psi(x) dx.$$

This is the average value obtained from many measurements on identically prepared systems.

7. A later application: the Hellmann–Feynman theorem

The definitions above are not just notation: they are exactly what we use to derive forces in ab-initio molecular dynamics. Let the Hamiltonian depend on the nuclear coordinates $\mathbf{R}$, and let $|\Psi_0(\mathbf{R})\rangle$ be an exact normalized eigenstate:

$$\hat{H}(\mathbf{R})|\Psi_0\rangle = E_0(\mathbf{R})|\Psi_0\rangle, \quad \langle\Psi_0|\Psi_0\rangle = 1.$$

The energy is

$$E_0 = \langle\Psi_0|\hat{H}|\Psi_0\rangle.$$

Differentiate with respect to the position of nucleus I :

$$\nabla_I E_0 = \langle\nabla_I\Psi_0|\hat{H}|\Psi_0\rangle + \langle\Psi_0|\nabla_I\hat{H}|\Psi_0\rangle + \langle\Psi_0|\hat{H}|\nabla_I\Psi_0\rangle.$$

Using the eigenvalue equation and Hermiticity, the first and third terms become

$$E_0 [\langle\nabla_I\Psi_0|\Psi_0\rangle + \langle\Psi_0|\nabla_I\Psi_0\rangle].$$

But differentiating the normalization condition gives

$$\nabla_I\langle\Psi_0|\Psi_0\rangle = 0,$$

so

$$\langle\nabla_I\Psi_0|\Psi_0\rangle + \langle\Psi_0|\nabla_I\Psi_0\rangle = 0.$$

The wavefunction-derivative terms therefore cancel as a sum, leaving

$$\nabla_I E_0 = \langle\Psi_0|\nabla_I\hat{H}|\Psi_0\rangle.$$

Hence the force is

$$\mathbf{F}_I = -\nabla_I E_0 = -\langle\Psi_0|\nabla_I\hat{H}|\Psi_0\rangle.$$

Why this is useful: although the electronic wavefunction changes when the nuclei move, for an exact normalized eigenstate we do not need to calculate that derivative explicitly in the Hellmann–Feynman contribution to the force.

Practical caveat: in numerical electronic-structure calculations, incomplete basis sets or incomplete electronic convergence can introduce additional force contributions/errors. In atom-centered basis sets, coordinate-dependent basis functions give rise to Pulay terms.

8. One-page memory map

Concept	Meaning / equation
Ket	$ \Psi\rangle$: the quantum state
Bra	$\langle\Psi = \Psi\rangle^\dagger$: Hermitian conjugate of the ket
Hermitian conjugate	transpose + complex conjugate
Complex conjugate	$(a + ib)^* = a - ib$
Inner product	$\langle\Phi \Psi\rangle = \int \Phi^* \Psi \, dx$
Normalization	$\langle\Psi \Psi\rangle = 1$
Orthogonality	$\langle\Psi_m \Psi_n\rangle = 0$ for distinct orthogonal states
Orthonormality	$\langle\Psi_m \Psi_n\rangle = \delta_{mn}$
Eigenvalue equation	$\hat{A} \Psi_n\rangle = a_n \Psi_n\rangle$
Hermitian operator	$\hat{A}^\dagger = \hat{A}$
Key consequences	Hermitian eigenvalues are real; different-eigenvalue eigenstates are orthogonal
Expectation value	$\langle A \rangle = \langle\Psi \hat{A} \Psi\rangle$
Probability density	$ \Psi ^2 = \Psi^* \Psi$

The goal is not to memorize every derivation, but to recognize the logic connecting states, operators, inner products, Hermiticity, and the equations used later in DFT and AIMD.